UNIVERSITY OF RWANDA COLLEGE OF SCIENCE AND TECHNOLOGY

**CAT 1 OPERATING SYSTEMS**

**QUESTION 1:**

a) Using Robin Round scheduling algorithm , create a Gantt Chart of processes and complete the table .Time quantum is 2 milliseconds **/ 2+2+2pts**

| Process | AT | BT | CT=? | AV TAT=? | AV.WT=? |
|---|---|---|---|---|---|
| P0 | 0 | 10 | | | |
| P1 | 1 | 6 | | | |
| P2 | 3 | 2 | | | |
| P3 | 5 | 4 | | | |

b) Discuss on operating systems services **/4pts**

**QUESTION 2:**

a) Distinguish the Asymmetric multiprocessing from Symmetric one **/2pts**
b) Explain different conditions for deadlock **/4pts**
c) Discuss on major functions of operating systems. **/4pts**

**QUESTION 3: / 4+3+3 pts**

Consider a set of processes with their arrival time and CPU burst Time and the Priority as given in the table. Calculate the Gantt chart , average TAT, and Average WT . The CPU priority preemptive scheduling algorithm should be applied. (zero priority is the higher)

| Process ID | Arrival Time | Bust time | Priority | CT =? | AV. TAT=? | AV. WT=? |
|---|---|---|---|---|---|---|
| P1 | 0 | 11 | 2 | | | |
| P2 | 5 | 28 | 0 | | | |
| P3 | 12 | 2 | 3 | | | |
| P4 | 2 | 10 | 1 | | | |
| P5 | 9 | 16 | 4 | | | |

**QUESTION 4:** Distinguish the following terminology in operating system:

a) Swapping from Paging **/3pts**
b) Deadlocking from semaphores **/2pts**
c) Aging from starvation **/2pts**
d) CPU schedulers **/3pts**

***Good Luck !!!***

CamScanner